



Program: Data Science

Course Title: MSDS 545 Introduction to Computational Software Engineering

Semester and Year: *Fall 2026*

Credit Hours: 3

Assignment 1: Intro to Git & GitHub

Instructions

Read the following information and complete the required deliverables.

In this lab, you will set up Git and GitHub and practice certain Git commands.

Part 1: Create a repository in GitHub

1. Create an account in GitHub.
 - a. <https://docs.github.com/en/get-started/start-your-journey/creating-an-account-on-github>
2. Create a repository by clicking on the “Repositories” tab, then click on “New Repository.”
 - a. Name your repository **username-git-starter** (replace username with your email username).
 - b. Keep the repository on “Private”
 - c. Check the box for “Add a README file”
 - d. Under “Add .gitignore”, select Python from the list.
 - e. Click “Create repository” and you’ve created your first repository!

Part 2: Configure Cloud Shell access to GitHub

1. Install Git

Windows:
<https://git-scm.com/install/windows>

Mac:
<https://git-scm.com/install/mac>
2. Configure Git using the following commands.

- a. Open PowerShell on Windows or Terminal on Mac.
- b. Run each of the following commands sequentially, and enter your corresponding GitHub username and email address used for GitHub:

```
git config --global user.name "<your name>"
git config --global user.email "<your email>@[studentemaildomain]"
git config --global init.defaultbranch main
```

3. Now, we need to create an SSH key pair. This is a way to connect GitHub and Cloud Shell.

- a. Copy and paste the following line into the PowerShell or Terminal
`ssh-keygen -t ed25519 -C "Cloud Shell"`
- b. ***. You will need to hit Enter 3 times to create a key without a passphrase. (press enter for each question)***

4. After running the above command:.

- a. The output should look something like: `ssh-ed25519`
`AAAAC3NzaC1lZDI1NTE5AAAAIDmWRDPfNPfrSRt7fG4rHpomXQZVv/hebVM`
`pc5/vpGTb Cloud Shell`

5. Now run the following command:

```
cat "$HOME/.ssh/id_ed25519.pub"
```

- a. ***Copy the ENTIRE output after this command is run.***

6. Add this SSH key [in your Github settings](#) and give it a title. By going through the following steps:

- a. Go to settings then go to SSH and GPG Keys
- b. Then click New SSH Key
- c. Then click add SSH Key
- d. For the title enter a title name and for the Key enter the copied output that was returned after running the cat command within the Powershell or Terminal window.

← → ↻ https://github.com

Relaunch to update

Dashboard

Q Type [Z] to search

asmahmuallem...

Set status

Profile

Repositories

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Try Enterprise **Free**

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asmahmuallem-mmc

Top repositories **New**

Find a repository...

MSDS-545/Assignment1_2

MSDS-545/food_access_app_multipleModels-

MSDS-545/cyber-ids-ml-demo

MSDS-545/CI-DEMO

MSDS-545/food_access_app

asmahmuallem-mmc/food_access_app_multiple_models

MSDS-545/vuln-scanner-dashboard

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Ask All repositories +

Auto

https://github.com/settings/profile

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asmahmuallem-mmc (asmahmuallem-mmc)

Your personal account

Switch settings context

Public profile

Account

Appearance

Accessibility

Notifications

Access

Billing and licensing

Emails

Password and authentication

Sessions

SSH and GPG keys

Credentials

Organizations

Enterprises

Teams

Moderation

Code, planning, and automation

Public profile

Name

Your name may appear around GitHub where you contribute or are mentioned. You can remove it at any time.

Public email

Select a verified email to display

You can manage verified email addresses in your [email settings](#)

Bio

Tell us a little bit about yourself

You can @mention other users and organizations to link to them.

Pronouns

Don't specify

URL

Social accounts

Profile picture

Edit

SSH keys

[New SSH key](#)

This is a list of SSH keys associated with your account. Remove any keys that you do not recognize.

Add new SSH Key

Title

Key type

Authentication Key ↕

Key

Begins with 'ssh-rsa', 'ecdsa-sha2-nistp256', 'ecdsa-sha2-nistp384', 'ecdsa-sha2-nistp521', 'ssh-ed25519', 'sk-ecdsa-sha2-nistp256@openssh.com', or 'sk-ssh-ed25519@openssh.com'

Add SSH key

This process is equivalent to “logging in” to GitHub. Now that the Cloud Shell has access, we need to tell it to use the repository you created by accepting the assignment in part 1.

Part 3: Cloning your repository

1. Download VS Code and user the instaler

Windows:

<https://code.visualstudio.com/docs/setup/windows>

See the [Download Visual Studio Code](#) page for the full list of installation options.

Install with the Windows installer

- 1 Download the [Visual Studio Code User setup](#)  for Windows.
- 2 Run the installer, `VSCodeUserSetup-{version}.exe`.

By default, the User setup installs VS Code under `C:\Users\{Username}\AppData\Local\Programs\Microsoft VS Code`.

Tip

Setup adds Visual Studio Code to your `%PATH%` environment variable

Mac:

<https://code.visualstudio.com/docs/setup/mac>

Installing Visual Studio Code on macOS

 Copy as Markdown 

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[{ Install VS Code on macOS](#)

Launch VS Code from the command line

Updates

Touch Bar support

After installation

Common questions

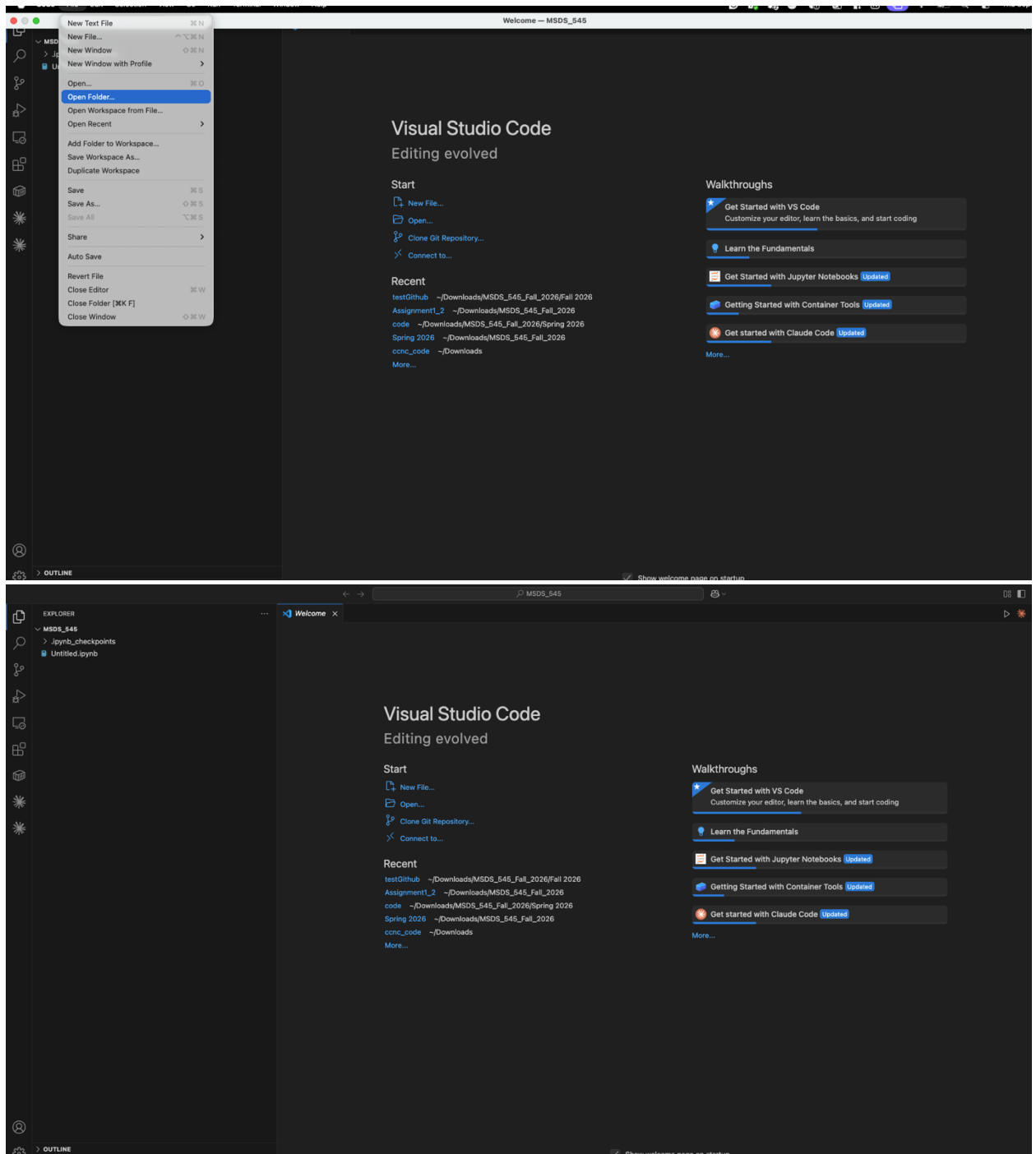
Visual Studio Code is available for macOS through a downloadable disk image. VS Code supports Intel-based Macs and Macs with Apple silicon.

Install VS Code on macOS

- 1 [Download Visual Studio Code](#)  for macOS.
- 2 Open the downloaded `.dmg` file.
- 3 Drag `Visual Studio Code.app` to the **Applications** folder.
- 4 Open VS Code from the **Applications** folder by double-clicking the icon.
- 5 To keep VS Code in the Dock, Control-click the icon in the Dock, then select **Options > Keep in Dock**.

Note

2. Create a folder on your local Drive MSDS_545
3. Now open VS Code and click on File Open folder, then naviage to the folder you created and open.



4. Close the Welcome window.
5. Go to your assignment (git-starter..) repository on GitHub.
6. Click on the green **Code** button and go to **Local** and select SSH option to copy the SSH cloning path:

Your main branch isn't protected

Dismiss

Protect this branch

Protect this branch from force pushing or deletion, or require status checks before merging. [View documentation.](#)

main

4 Branches

0 Tags

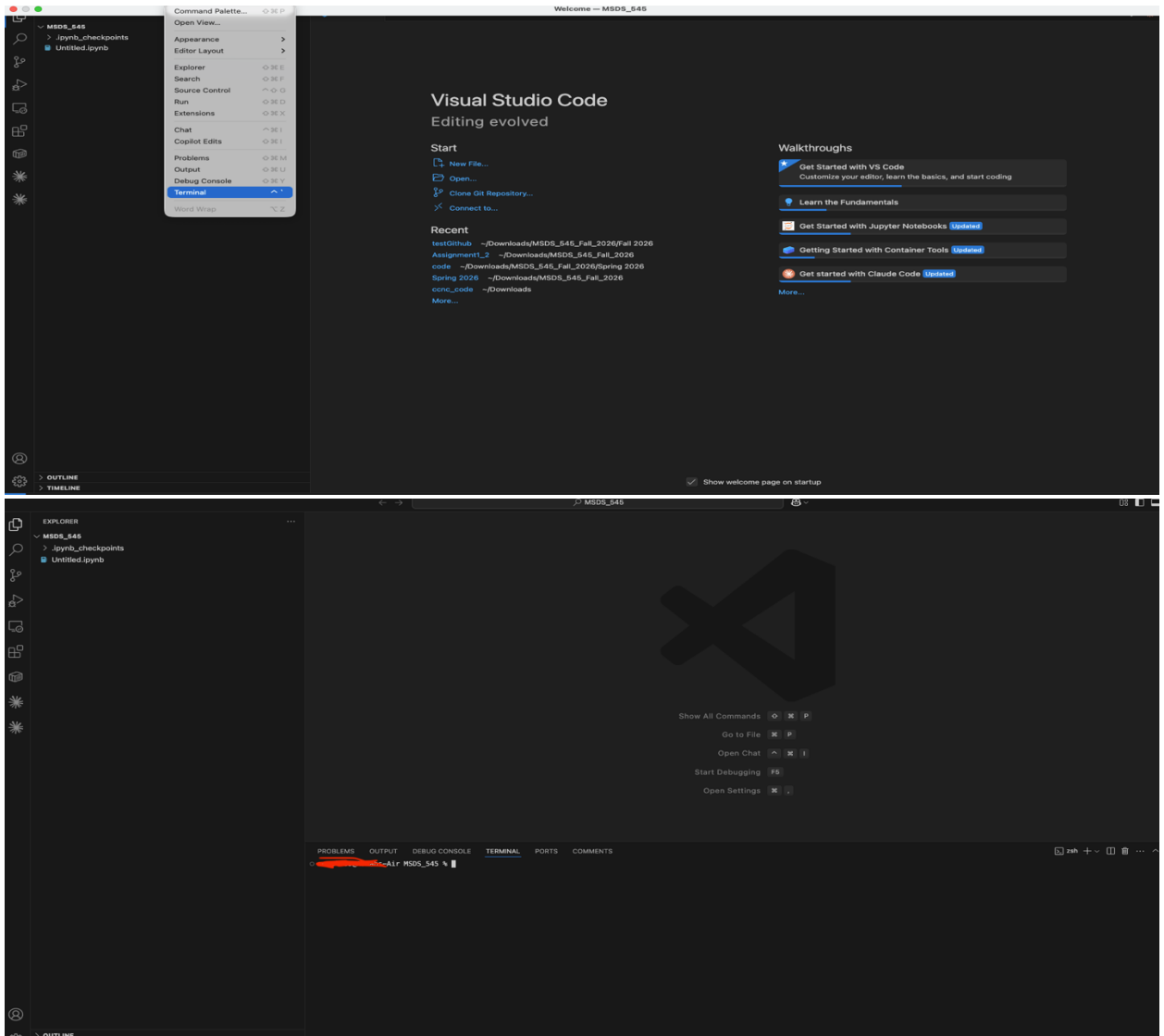
Go to file

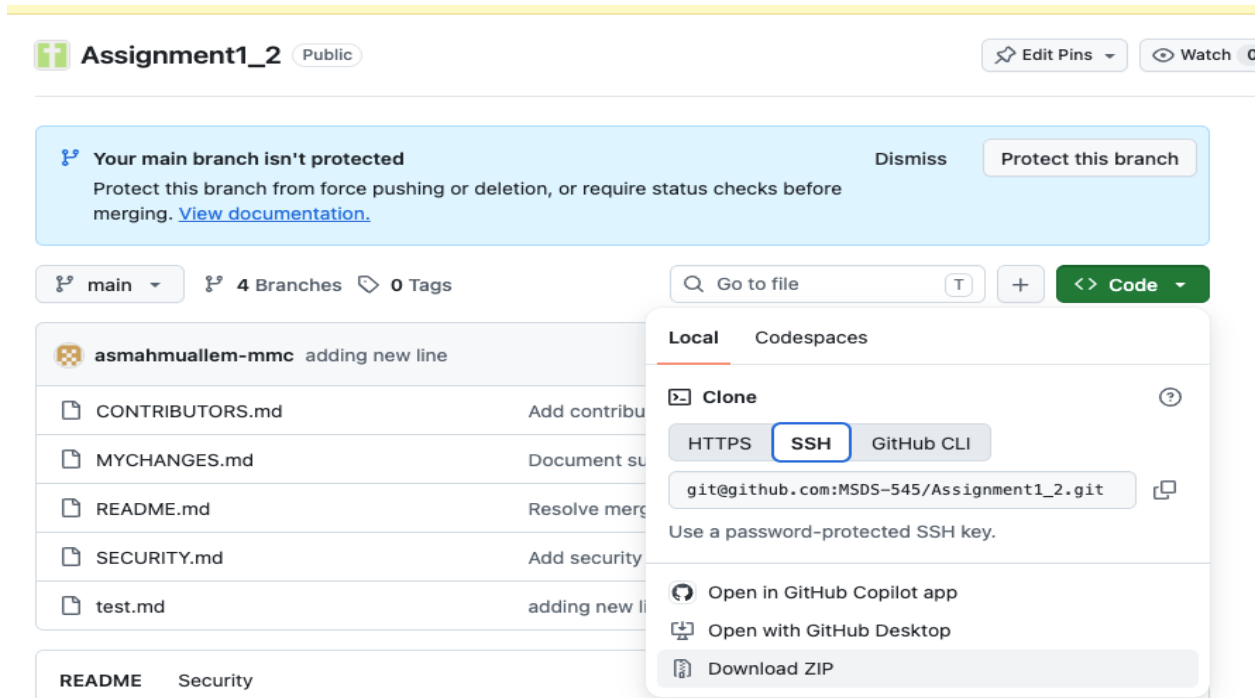
T

+

<> Code

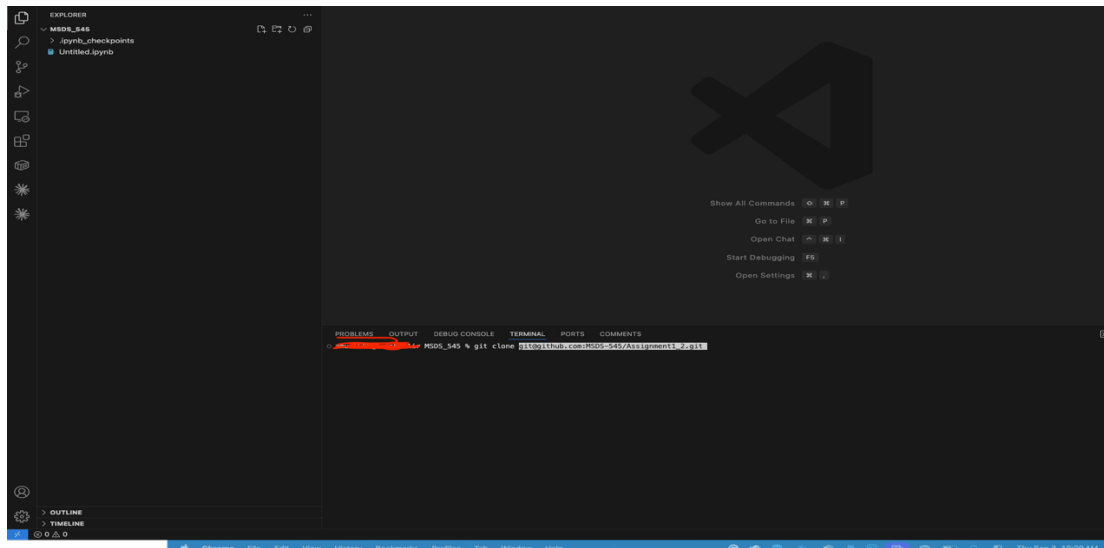
- After that go to the terminal in VS Code, once you open, you will see a terminal window at the bottom of the screen:





8. In the terminal window type the following command

`git clone (repository name from github copied ssh)`



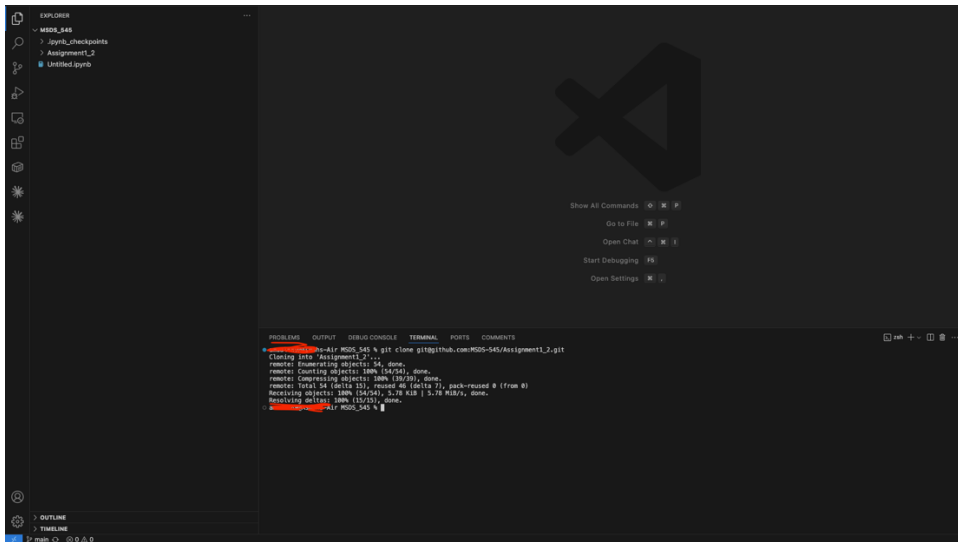
9. Press enter after running the command, you will see the following:

The authenticity of host 'github.com (192.30.255.112)' can't be established.

ECDSA key fingerprint is SHA256:<long string>.

Are you sure you want to continue connecting (yes/no/[fingerprint])?

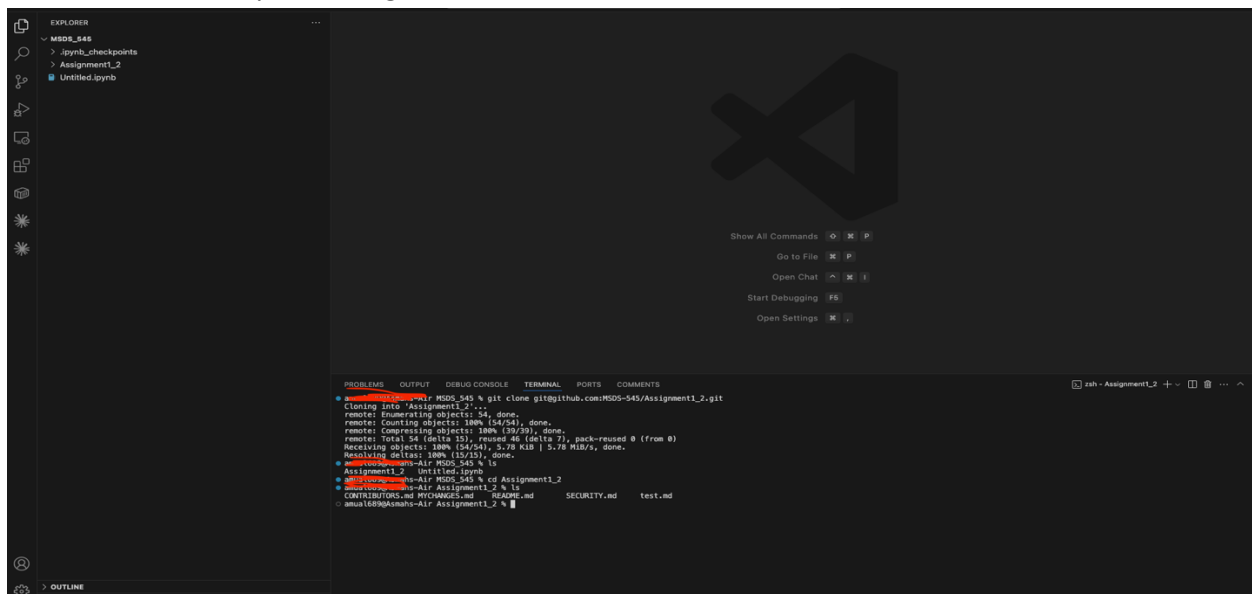
10. Enter yes and press enter.



11. Run `ls` (list files) on the command line after accepting the connection. What do you see?

Record your findings in the lab form.

12. Change into the new directory that was cloned using `cd` and run `ls` again. Now what do you see? Record your findings in the lab form.

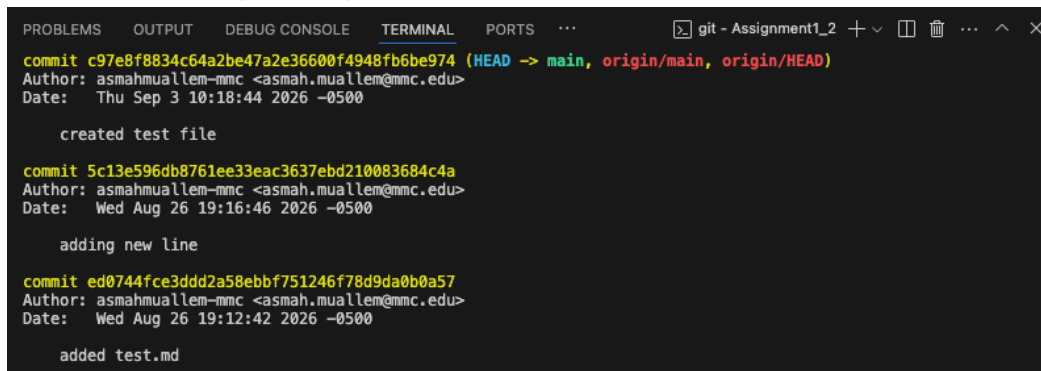


Part 4: Using Git

This section introduces the most common and helpful informational git commands: `git status` and `git log`.

1. From the terminal, navigate inside of your assignment repository. I.e. running `ls` should show your starter files.
2. Run `git status`. Record your output and what it means in the lab form.
3. Run `git log` to see the previous commits. Record your output and what it means in the lab form.

4. Press **q** to exit the **git log** interface.



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS ... git - Assignment1_2 + - - - - -
commit c97e8f8834c64a2be47a2e36600f4948fb6be974 (HEAD -> main, origin/main, origin/HEAD)
Author: asmahmuallem-mmc <asmah.muallem@mmc.edu>
Date: Thu Sep 3 10:18:44 2026 -0500

    created test file

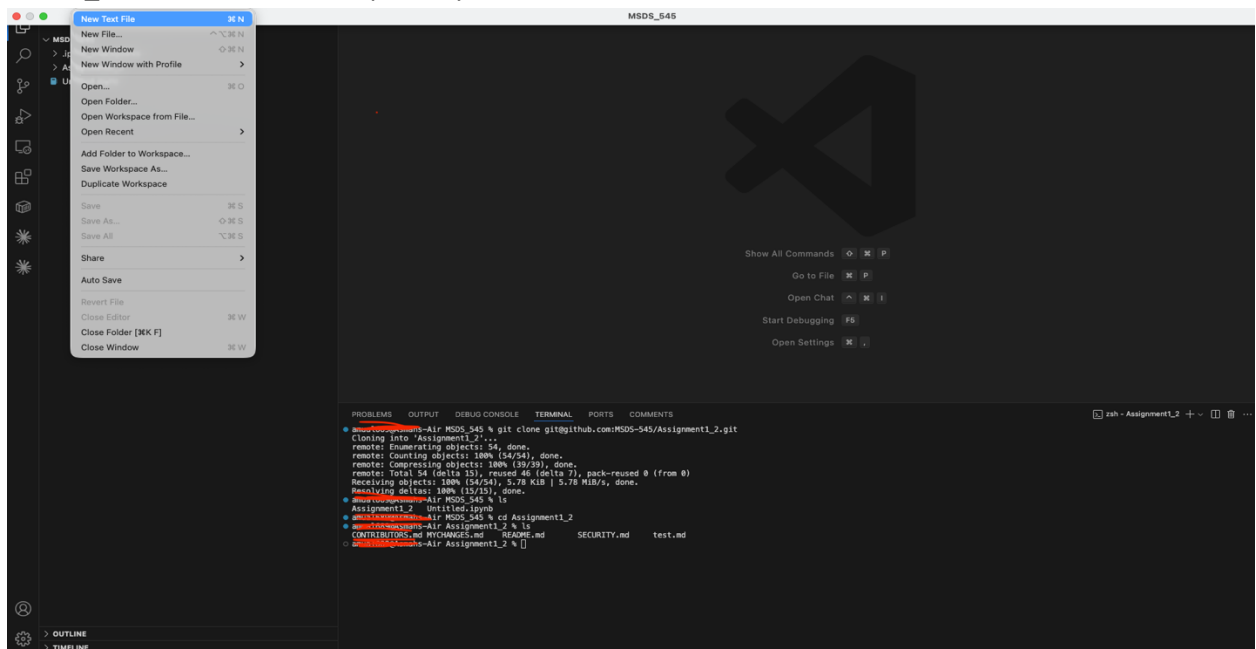
commit 5c13e596db8761ee33eac3637ebd210083684c4a
Author: asmahmuallem-mmc <asmah.muallem@mmc.edu>
Date: Wed Aug 26 19:16:46 2026 -0500

    adding new line

commit ed0744fce3ddd2a58ebbf751246f78d9da0b0a57
Author: asmahmuallem-mmc <asmah.muallem@mmc.edu>
Date: Wed Aug 26 19:12:42 2026 -0500

    added test.md
```

5. Try adding a file and committing.
6. Click New Text File, then add text to the file as shown in the snippet. Following this, save the file as `test_file.txt` in the cloned repository.



7. Now go to the terminal and run **git status**. Record your output and what it means in the lab form.

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS COMMENTS
remote: Enumerating objects: 54, done.
remote: Counting objects: 100% (54/54), done.
remote: Compressing objects: 100% (39/39), done.
remote: Total 54 (delta 15), reused 46 (delta 7), pack-reused 0 (from 0)
Receiving objects: 100% (54/54), 5.78 KiB | 5.78 MiB/s, done.
Resolving deltas: 100% (15/15), done.
amuel689@asmahs-Air MSDS_545 % ls
Assignment1_2 Untitled.ipynb
amuel689@asmahs-Air MSDS_545 % cd Assignment1_2
amuel689@asmahs-Air Assignment1_2 % ls
CONTRIBUTORS.md MYCHANGES.md README.md SECURITY.md test.md
amuel689@asmahs-Air Assignment1_2 % git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
amuel689@asmahs-Air Assignment1_2 % git status
On branch main
Your branch is up to date with 'origin/main'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    Test file.txt

nothing added to commit but untracked files present (use "git add" to track)
amuel689@asmahs-Air Assignment1_2 %
```

8. Now run `git add <file name>` and run `git status`. Record your output and what it means in the lab form.

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS COMMENTS
amuel689@asmahs-Air Assignment1_2 % ls
CONTRIBUTORS.md MYCHANGES.md README.md SECURITY.md test.md
amuel689@asmahs-Air Assignment1_2 % git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
amuel689@asmahs-Air Assignment1_2 % git status
On branch main
Your branch is up to date with 'origin/main'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    Test file.txt

nothing added to commit but untracked files present (use "git add" to track)
amuel689@asmahs-Air Assignment1_2 % git add Test\ file.txt
amuel689@asmahs-Air Assignment1_2 % git status
On branch main
Your branch is up to date with 'origin/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    new file:   Test file.txt

amuel689@asmahs-Air Assignment1_2 %
```

9. Now run `git commit -m "message"` and give a message. Record your output and what it means in the lab form.
10. Now run `git push`. Record your output and what it means in the lab form.

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS COMMENTS
● amual689@Asmahs-Air Assignment1_2 % ls
CONTRIBUTORS.md MYCHANGES.md README.md SECURITY.md test.md
● amual689@Asmahs-Air Assignment1_2 % git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
● amual689@Asmahs-Air Assignment1_2 % git status
On branch main
Your branch is up to date with 'origin/main'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    Test file.txt

nothing added to commit but untracked files present (use "git add" to track)
● amual689@Asmahs-Air Assignment1_2 % git add Test\ file.txt
● amual689@Asmahs-Air Assignment1_2 % git status
On branch main
Your branch is up to date with 'origin/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    new file:   Test file.txt
○ amual689@Asmahs-Air Assignment1_2 %
```

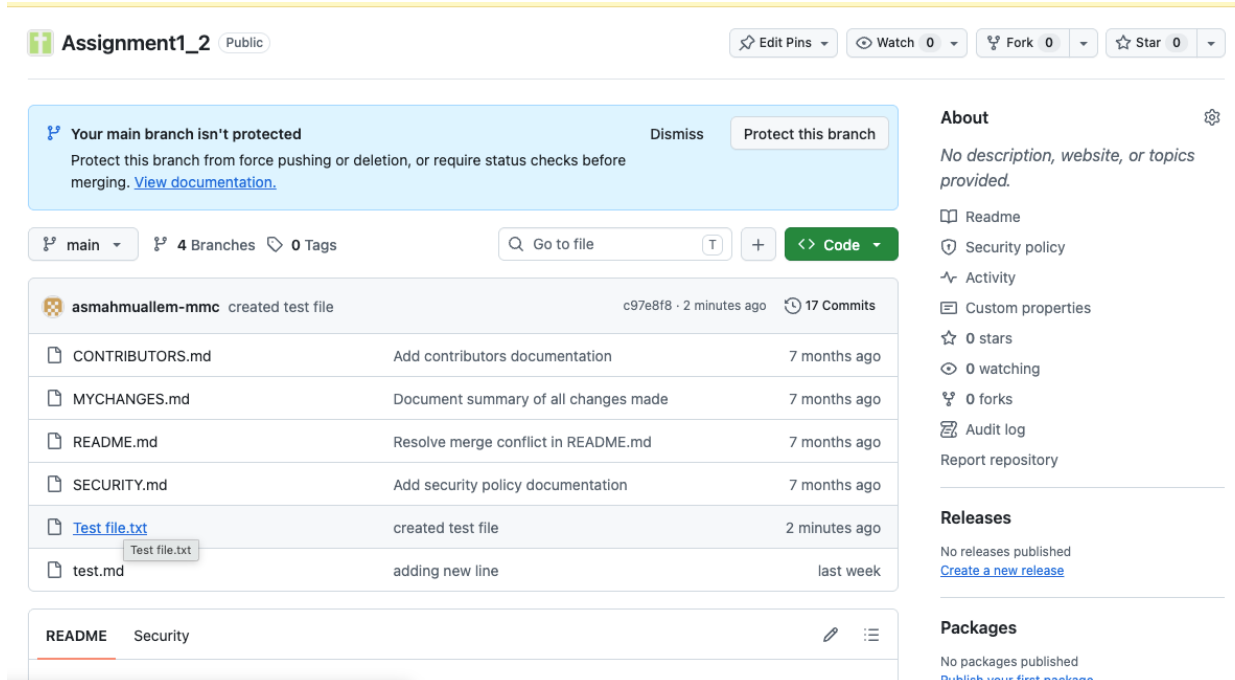
```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS COMMENTS

nothing added to commit but untracked files present (use "git add" to track)
● amual689@Asmahs-Air Assignment1_2 % git add Test\ file.txt
● amual689@Asmahs-Air Assignment1_2 % git status
On branch main
Your branch is up to date with 'origin/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    new file:   Test file.txt

● amual689@Asmahs-Air Assignment1_2 % git commit -m "created test file"
[main c97e8f8] created test file
 1 file changed, 1 insertion(+)
 create mode 100644 Test file.txt
● amual689@Asmahs-Air Assignment1_2 % git push
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 283 bytes | 283.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:MSDS-545/Assignment1_2.git
 5c13e59..c97e8f8  main -> main
○ amual689@Asmahs-Air Assignment1_2 %
```

11. Check the newly added file in Github.



Optional

Go to the following page and play the git game:

Use <https://learngitbranching.js.org/> to get familiar with Git. Level up by completing:

- Main > Introduction Sequence
 - 1: Introduction to Git Commits
 - 2: Branching in Git
 - 3: Merging in Git
- Remote > Push & Pull
 - 3: Git Fetchin'
 - 4: Git Pullin'
 - 5: Faking Teamwork
 - 6: Git Pushin'

Create a reference guide in a text or markdown file and submit it in your repository with the filename `git_reference.txt` or `git_reference.md`. You can use the github UI to do this or create a file in the terminal using vi command, or through the github desktop add a file with this name to your repository.

Within the file, add how many levels you passed!

Deliverables:

- Upload your Git reference guide to your Git repository.
- Share the github repository in a document file and upload to the BB portal with the assignment submission along with the file from Lab 2_2.
- Include snippets of all the steps.

Resources:

- [Learn Git Branching](#) (Game)
- [Git Immersion](#) (Friendly lab-like docs)
- [Git HowTo](#) (Friendly lab-like docs)
- [Git - Reference](#) (Source documentation)